TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-driven design and manufacturing of high-performance porous electrodes for electrochemical energy systems			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical Materials		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Metal materials	Materials Analysis/Evaluation Technologies	R&D/Engineering Services	Testing/Inspection/Analysis Techniques
Concept of Proposed Technology	The project establishes a data- and model-driven design and manufacturing platform for porous electrodes used in electrolyzers and related electrochemical devices. It couples materials informatics, physics-informed machine learning, and design-of-experiments with cross-scale diagnostics (from microstructure and interfaces up to full cells).			
Rationale for Joint R&D Project	Korean SME targets rapid scale-up of advanced electrodes for Korean energy markets. Fraunhofer IMWS contributes cross-scale diagnostics, microstructure analytics for materials and processes. Joint R&D reduces development risk and cycle time, raises yield and reliability, and delivers AI-based electrode designs and process guidelines, advancing technology readiness from laboratory verification to pilot validation.			
Key Words	AI-driven electrode design; porous electrode microstructure; mass transport and catalyst utilization;			

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	- Build a data- and model-driven design platform for porous electrodes, linking composition, microstructure, and process parameters to performance and durability. - Optimize interfacial adhesion to improve catalyst utilization, mass transport, and area-specific resistance. - Implement cross-scale diagnostics and microstructure segmentation to quantify critical features (porosity, tortuosity, crack density, interface quality) and correlate them with operational metrics.			
R&D Target Specifications	- Structure–property targets: porosity and tortuosity tuned for target mass transport; interfacial adhesion above defined thresholds; minimized area-specific resistance; controlled thickness and uniformity - Operational targets: stable performance under relevant operating envelopes (current density, temperature, pressure, humidity) with defined degradation-rate limits and repeatability metrics. - Data/AI targets: curated datasets, feature extraction from microstructure images, predictive models with validated accuracy and uncertainty bounds; decision support for process tuning.			
R&D Plan	Phase 1 – define product scope, acceptance criteria, and operating envelopes; set up data pipeline and measurement protocols; Phase 2 – Data and diagnostics: expand datasets; implement microstructure segmentation and feature extraction; establish process–structure–property maps; Phase 3 – Model development and process optimization: build physics-informed and machine-learning models; optimize formulation,			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- Lead cross-scale diagnostics (microstructure, interfaces, full-cell correlations) and develop AI/physics-informed models for process–structure–property relations. - Establish data pipelines, feature extraction, and segmentation workflows; define test protocols, acceptance criteria, and quality gates. - Support formulation and process optimization; validate performance and durability;
Expected R&R of Korean SME Partner	Provide materials, electrode concepts, and production constraints; fabricate prototypes and pilot batches. - Execute design-of-experiments and integrate process changes on the manufacturing line; supply operational test data and feedback. - Lead commercialization and rollout in Korea; ensure on-site deployment, training, and maintenance for pilot-scale adoption.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	- Validated electrode architectures and manufacturing workflows that meet defined performance, durability, and yield targets. - Data- and model-driven design toolkit (datasets, feature extraction, predictive models) enabling faster iteration and reduced development risk. - Pilot-scale demonstration (TRL 6) with documented process windows, acceptance criteria, and transfer package for replication across products - Enhanced competitiveness of Neel Science in Korean energy markets through reliable, scalable electrode production and shortened time-to-market.

6. References

(1)

(2)

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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